

Engineering review and document authority

This report combines verified frozen active-section geometry with the current system-owned end-design basis. End geometry is not selected. It is a review document, not a fabrication release or a completed pressure-vessel design.

Scope	Disposition
Frozen active section	Ø700 mm; 20 compartments; active height 4200 mm. Unchanged.
Current ends	SYSTEM PENDING. Diameters, residence heights, opening envelopes and overall elevations are undetermined.
Five engineering drawings	Current GA, active-only longitudinal section, typical compartment, rotor and stator.
Historical end/nozzle reservations	Not current authority. Excluded from the current schedules and longitudinal section.
Calculation & Audit Archive	Separate complete saved report: detailed source records, coordinates, validation, provenance and historical reservations.

Tables use engineering display rounding (normally 0.01 mm); no manufacturing tolerance is implied. Drawings retain their existing saved-component precision. Page count follows meaningful content, not a target length.

Frozen active design basis

Quantity	Frozen value
Column internal diameter	700 mm
Compartment pitch	210 mm
Physical compartment count	20
Installed active height	4200 mm
Required active height	4200 mm
Rotor diameter	231 mm
Rotor / column ratio	0.33
Stator free-area input	0.4
Selected speed	30 rpm
Inherited speed bounds	30 to 30 rpm
Phase configuration	rrbo-continuous-nmp-dispersed
Sizing method	ADOPTED_COMPARTMENT_EFFICIENCY
Design theoretical stages	7
Compartment efficiency (if adopted)	0.35

An equal lower/upper speed is a selected point, not an established operating window. No upstream hydraulic or stage-efficiency policy is changed by this report.

Process-flow directions and authority separation

Basis	Meaning
Saved flow arrangement	nmp-down-rrbo-up
Flow classification	System-generated
Normal solvent/oil mass ratio	0.6
Nozzle-only solvent/oil mass ratio	1.5 MASS; never a product redistribution or residence basis
Normal product duty	NORMAL_PRODUCT_AUTHORITY_PENDING

Saved arrangement: RRBO/oil enters at the lower end and travels upward toward raffinate withdrawal; wet solvent enters at the upper end and travels downward toward extract withdrawal.

Direction does not qualify interface levels, connection elevations or outlet allocation. Historical connection coordinates are not substituted for current pending locations.

Active internals — dimensional schedule

Explicit active-component values only. The shaft diameter is retained; full shaft length, supports, seal and drive elevations depend on unresolved current end arrangements and are not republished as current.

Frozen parameter	Value	Unit
column Diameter	700	mm
installed Active Height	4200	mm
compartment Height	210	mm
rotor Diameter	231	mm
shaft Diameter	44	mm
stator Thickness	4	mm
stator Opening Diameter	112	mm
stator Radial Width	294	mm
blade Count	6	count
blade Height	28	mm
blade Thickness	3	mm
blade Radial Length	80.5	mm
blade Inner Segment Length	30	mm
blade Outer Segment Length	50.5	mm
blade Inner Segment Height	16	mm
blade Outer Segment Height	28	mm
hub Diameter	70	mm
hub Height	24	mm
rotor Offset	105	mm
shroud Inner Diameter	130	mm
shroud Outer Diameter	231	mm
shroud Thickness	2	mm
stator Hole Diameter	46.73	mm

Active internals — dimensional schedule (continued)

Frozen parameter	Value	Unit
stator Lane Width	8	mm
column Area area	0.3848	m ²
shaft Area area	0.0021	m ²
stator Gross Open Area area	0.1539	m ²
stator Net Open Area area	0.1524	m ²

Rotor and stator construction

Component	Frozen construction
Rotor	Preliminary double-entry shrouded six-blade turbine; retained 130mm eye
Stator	Preliminary perforated stator; retained 112mm centre and 84-hole architecture
Free-area convention	Gross physical area: $(0.112^2 + 84 \times 0.046733285782191686^2) / 0.7^2 = 0.4$. Shaft-corrected diagnostic separate; no hydraulic equivalence asserted.
Rotor hub diameter / height	70 / 24 mm
Rotor swept diameter	231 mm
Blade azimuths	0, 60, 120, 180, 240, 300 degrees
Shaft diameter	44 mm

Frozen active component	Quantity	Saved diameter [mm]
Rotor assemblies	20	231
Stator plates (including active boundaries)	21	700

Stator perforation ring	PCD [mm]	Hole count
1	211.47	12
2	315.2	18
3	420	24
4	530	30

Saved perforated stator: 84 holes, diameter 46.73 mm. Exact saved hole centres remain in the detail drawing and archive; no coordinate dump is repeated here.

Component profiles and perforations are rendered through the existing frozen-detail renderer, not recreated from schematic approximations. Saved adapted/preliminary component qualifications are unchanged.

Key geometric clearances and calculation summary

Geometric diagnostic	Calculation	Result [mm]
Rotor tip to column wall (radial)	$(D - \text{rotor swept diameter}) / 2$	234.5
Shaft to stator central opening (radial)	$(\text{opening diameter} - \text{shaft diameter}) / 2$	34
Pitch less stator thickness	$\text{pitch} - \text{stator plate thickness}$	206
Nominal axial half-clearance	$(\text{pitch} - \text{stator thickness} - \text{rotor axial envelope}) / 2$	87

These are presentation diagnostics, not allowable tolerances or dynamic shaft-clearance checks. The axial diagnostic uses the complete saved rotor axial envelope (32 mm), including hub and shrouds when present, centred in the compartment. Deflection and assembly tolerances still require mechanical assessment. Detailed component profiles govern.

Stack check: $20 \times 210 \text{ mm} = 4200 \text{ mm}$; inherited installed active height 4200 mm.

Compartment elevations — relative active datum

All elevations are relative to the frozen active bottom = 0 mm. A datum translation only: original internals and pitch are unchanged. No vessel-bottom, head or overall equipment datum is implied.

Compartment	Bottom [mm]	Rotor [mm]	Stator [mm]	Top [mm]
1	0	105	210	210
2	210	315	420	420
3	420	525	630	630
4	630	735	840	840
5	840	945	1050	1050
6	1050	1155	1260	1260
7	1260	1365	1470	1470
8	1470	1575	1680	1680
9	1680	1785	1890	1890
10	1890	1995	2100	2100
11	2100	2205	2310	2310
12	2310	2415	2520	2520
13	2520	2625	2730	2730
14	2730	2835	2940	2940
15	2940	3045	3150	3150
16	3150	3255	3360	3360
17	3360	3465	3570	3570
18	3570	3675	3780	3780
19	3780	3885	3990	3990
20	3990	4095	4200	4200

Normal process material basis

Quantity	Normal basis
Oil volumetric feed	4 m ³ /h
Oil density	869 kg/m ³
Oil mass rate	3476 kg/h
Saved normal S/O mass ratio	0.6
Normal wet-solvent mass rate	2085.6 kg/h
Wet-solvent feed density	1015 kg/m ³
Normal wet-solvent volume rate	2.0548 m ³ /h
Combined normal feed	5561.6 kg/h

Component	Normal feed [kg/h]
saturates	2954.6
monoAromatics	243.32
diAromatics	139.04
polyAromatics	69.52
polarAromatics	69.52
nmp	2075.17
water	10.43

Normal solvent mass = saved S/O × oil mass. Feed phase densities are not qualified product densities; no product allocation is inferred from these totals.

Current end-section design basis

Frozen Stage 4 has no bound Stage 2 product reference. Normal product flows remain unknown; nozzle-only S/O does not resolve or cause this hold.

Requirement	Current engineering disposition
Source-qualified normal product duty	NORMAL_PRODUCT_AUTHORITY_PENDING
Independent diameter-selection rule	NO_ADMITTED_SYSTEM_DIAMETER_RULE
Top / bottom shell diameter	TBD / TBD — no automatic numerical selection admitted
Residence height / straight shell / total assembly	TBD — no numerical geometry fabricated
Additional feed-distribution necks	Ø700 mm outside frozen active height; lengths pending
Transitions	Conical portion 30°; physical knuckles and lengths pending; minimum max(0.200 m, 0.4D)
Interfaces	±0.150 m from transitions, not an overall vessel elevation
Opening envelope	Unknown; nozzle centre and near/far opening edges unqualified
Extension after opening	From far opening edge: minimum max(0.200 m, 0.4D); D pending
Heads	Current symbolic torispherical intent only; mechanical profiles pending; zero residence credit

Residence calculation and closure holds

Ten-minute hold-up: $V_{10} = Q_{\text{normal}} / 6$, with Q_{normal} in m^3/h . Usable straight volume = $0.90 \times (\pi D^2/4) \times H_{10}$. Therefore $H_{10} = Q_{\text{normal}} / [6 \times 0.90 \times (\pi D^2/4)]$.

This is one relationship between diameter and height. It does not uniquely choose both, even when normal flow and product density are qualified. No economic optimum, aspect ratio, maximum height or arbitrary candidate diameter is introduced.

Duty	Qualified volume [m^3/h]	H10 [m]
Top normal product	TBD	TBD — D unselected
Bottom normal product	TBD	TBD — D unselected

- Source-qualified NORMAL raffinate/extract flows and operating-temperature densities.
- An approved independent system diameter-selection rule. Ten-minute residence relates diameter and height; it does not uniquely select either.
- Qualified opening envelopes and mechanical neck, transition and head geometry.

Residence extends from the interface to the near product-opening edge. The post-opening extension starts at the far edge. Heads receive no residence-volume credit. Nozzle-only S/O must not enter this calculation.

Current preliminary nozzle schedule

Service	Basis m ³ /h	120% m ³ /h	Screen DN	Bore mm	v120 m/s
Oil feed	4	4.8	DN65	62.68	0.432
Wet-solvent feed	5.1369	6.1643	DN80	77.92	0.359
Raffinate outlet	TBD	TBD	TBD	TBD	TBD
Extract outlet	TBD	TBD	TBD	TBD	TBD

Hydraulic screen assumption: maximum velocity 0.5 m/s at 120% basis flow. Exact bore = outside diameter – 2 × wall. Provisional DN is the first passing candidate in the service-returned pipe table, not a nominal-bore substitution.

NOZZLES ONLY: solvent mass = $1.5 \times 3476 = 5214$ kg/h. Solvent volume = that mass / saved wet-solvent density = 5.1369 m³/h.

No blanket example DN is imposed. Outlet envelopes, branch loads, pressure rating, materials, reinforcement, projection and all current nozzle positions remain pending; historical nozzle bores and positions are excluded.

Oil feed: hydraulic pipe screen

$Q_{\text{basis}} = 4 \text{ m}^3/\text{h}$. $Q_{\text{check}} = 1.20 Q_{\text{basis}} = 4.8 \text{ m}^3/\text{h}$. Required bore = $\sqrt{[4 Q_{\text{check}} / (3600 \pi \times 0.5)]} = 58.27 \text{ mm}$.

DN	OD mm	Wall mm	Bore mm	v120 m/s	≤0.5
40	48.3	3.68	40.94	1.013	FAIL
50	60.3	3.91	52.48	0.616	FAIL
65	73	5.16	62.68	0.432	PASS
80	88.9	5.49	77.92	0.28	PASS
100	114.3	6.02	102.26	0.162	PASS

Service-returned preliminary selection: DN65.

Velocity = $(1.20 \times Q_{\text{basis}} / 3600) / [\pi \times (\text{bore}/1000)^2 / 4]$. Pipe OD and wall values are screen candidates, not approved material/rating selections. This screen does not qualify any product duty or vessel opening envelope.

Wet-solvent feed — nozzle-only S/O: hydraulic pipe screen

$Q_{\text{basis}} = 5.1369 \text{ m}^3/\text{h}$. $Q_{\text{check}} = 1.20 Q_{\text{basis}} = 6.1643 \text{ m}^3/\text{h}$. Required bore = $\sqrt{[4 Q_{\text{check}} / (3600 \pi \times 0.5)]} = 66.03 \text{ mm}$.

DN	OD mm	Wall mm	Bore mm	v120 m/s	≤0.5
40	48.3	3.68	40.94	1.301	FAIL
50	60.3	3.91	52.48	0.792	FAIL
65	73	5.16	62.68	0.555	FAIL
80	88.9	5.49	77.92	0.359	PASS
100	114.3	6.02	102.26	0.208	PASS

Service-returned preliminary selection: DN80.

Velocity = $(1.20 \times Q_{\text{basis}} / 3600) / [\pi \times (\text{bore}/1000)^2 / 4]$. Pipe OD and wall values are screen candidates, not approved material/rating selections. This screen does not qualify any product duty or vessel opening envelope.

Design validation summary and outstanding work

Saved validation register	Count
PASS	305
FAIL	0
TBD	0

Counts describe the complete historical frozen construction register, including historical end reservations; they do not validate the current pending ends. Individual machine check rows and source calculations are retained in the Audit Archive.

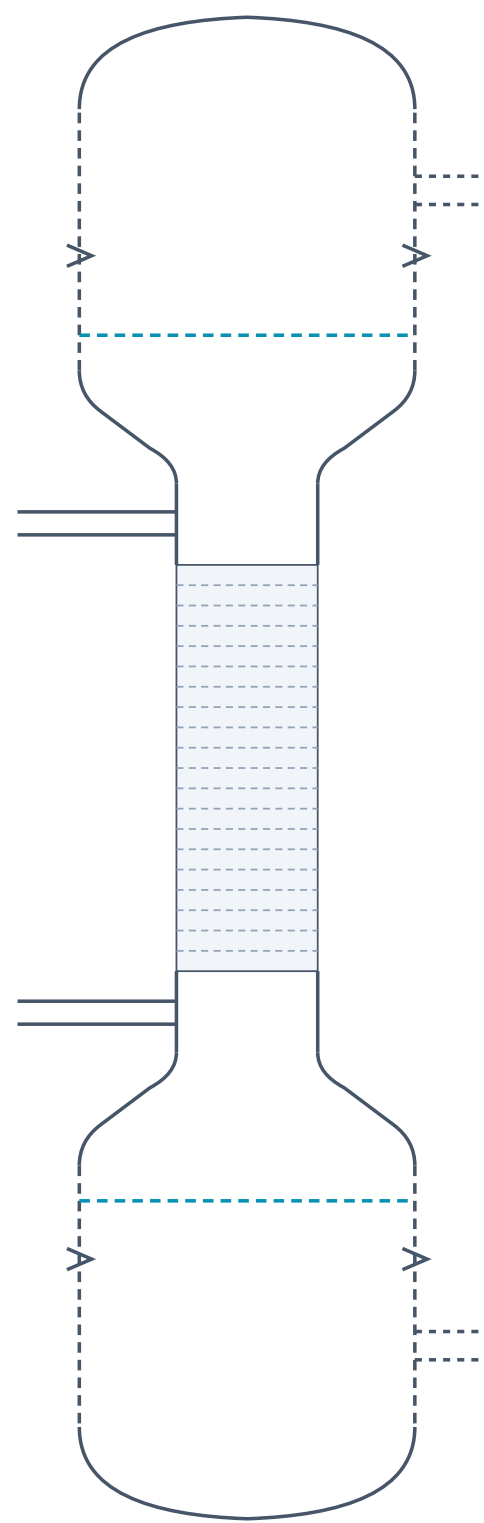
Engineering area	Release condition
Normal end-product duty	Designated simultaneous source-qualified raffinate/extract flow and operating-temperature density
End geometry	Independent diameter rule, opening envelope and mechanical transition/head closure
Rotating assembly	Strength, deflection, critical speed, tolerances, bearings, seals and drive assessment
Pressure equipment	Design pressure/temperature, materials, wall/head thickness and reinforcement
Interfaces and installation	Loads, supports, nozzle positions, maintainability and overall elevations

Frozen geometric completeness is not fabrication readiness, mechanical adequacy, product duty qualification or completed end sizing. No approval is inferred from a historical PASS.

General arrangement — current symbolic ends

CURRENT SYSTEM ARRANGEMENT — END DIMENSIONS PENDING

NOT TO SCALE. End widths/heights are symbolic, not chosen diameters or physical dimensions.
NORMAL process S/O=0.6 mass. S/O=1.5 MASS FOR NOZZLES ONLY.



TOP / RAFFINATE — SYSTEM DESIGN PENDING
Head profile symbolic; depth/radii TBD; zero residence credit
Beyond far edge: extension ≥ max(0.200 m, 0.4D); D pending
Product opening: near/far edges and nozzle envelope pending
Shell diameter and H10 unresolved — no selected dimensions
10 min NORMAL flow / 0.90 usable straight volume only
Interface rule: 0.150 m above transition

30° conical portion; formed junctions and physical length TBD
Transition ≥ max(0.200 m, 0.4D); no numeric D admitted

Additional Ø700 neck outside active: wet solvent P03

FROZEN ACTIVE SECTION — UNCHANGED
Ø700 mm × 20 compartments; active height 4.200 m
Revision 8; inherited internals remain in frozen detail tabs.
No user diameter comparison or historical preference governs this GA.
No overall height or absolute outlet elevations established.

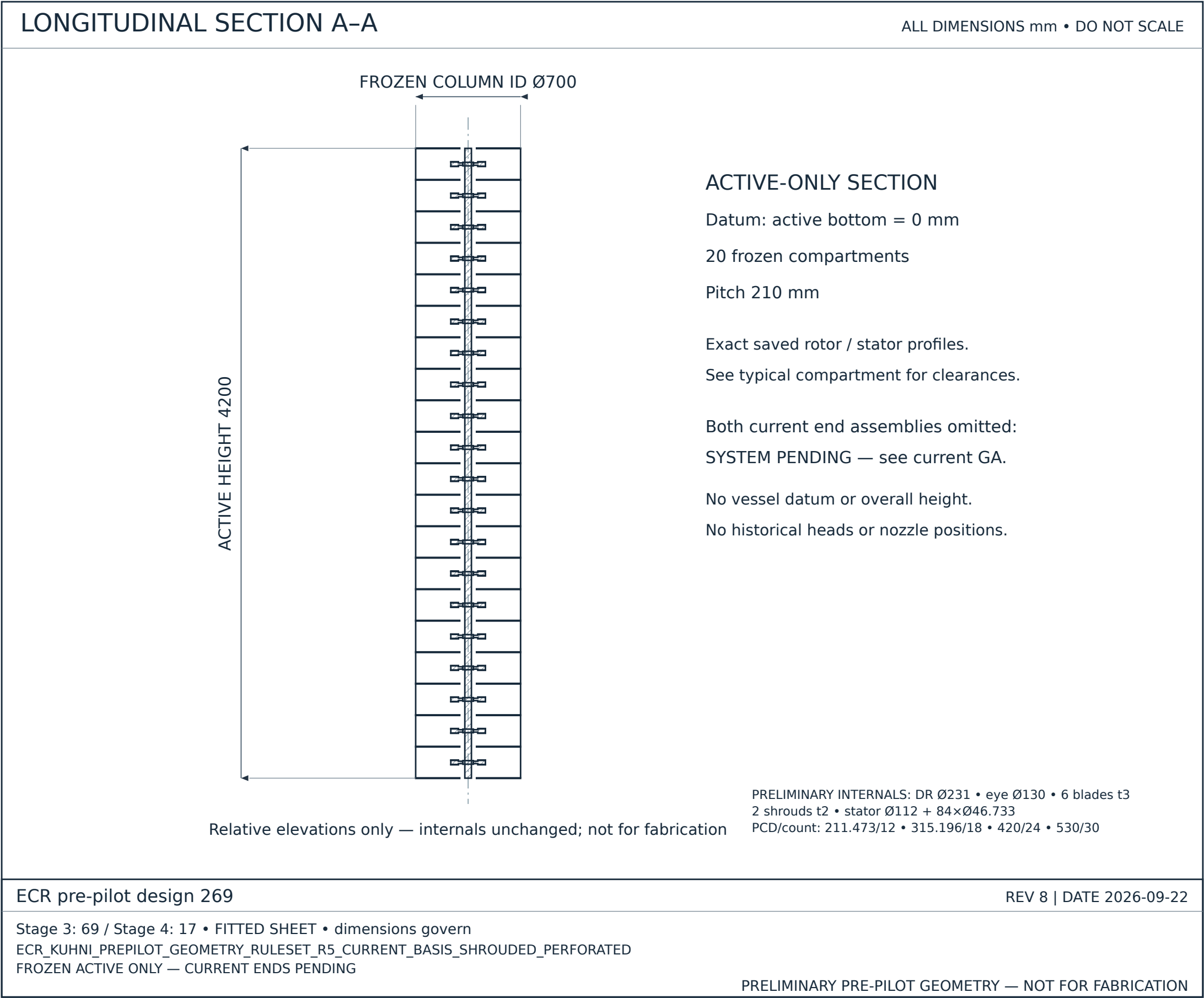
Additional Ø700 neck outside active: oil feed P01

Transition ≥ max(0.200 m, 0.4D); no numeric D admitted
30° conical portion; formed junctions and physical length TBD

Interface rule: 0.150 m below transition
10 min NORMAL flow / 0.90 usable straight volume only
Shell diameter and H10 unresolved — no selected dimensions
Product opening: near/far edges and nozzle envelope pending
Beyond far edge: extension ≥ max(0.200 m, 0.4D); D pending
Head profile symbolic; depth/radii TBD; zero residence credit
BOTTOM / EXTRACT — SYSTEM DESIGN PENDING

Required: source-qualified normal product duty and an independent system diameter-selection rule.
Ten-minute residence alone cannot uniquely determine both diameter and height.
Opening envelopes, necks, formed junctions and heads need mechanical authority. NOT FOR FABRICATION.

Longitudinal section — frozen active only



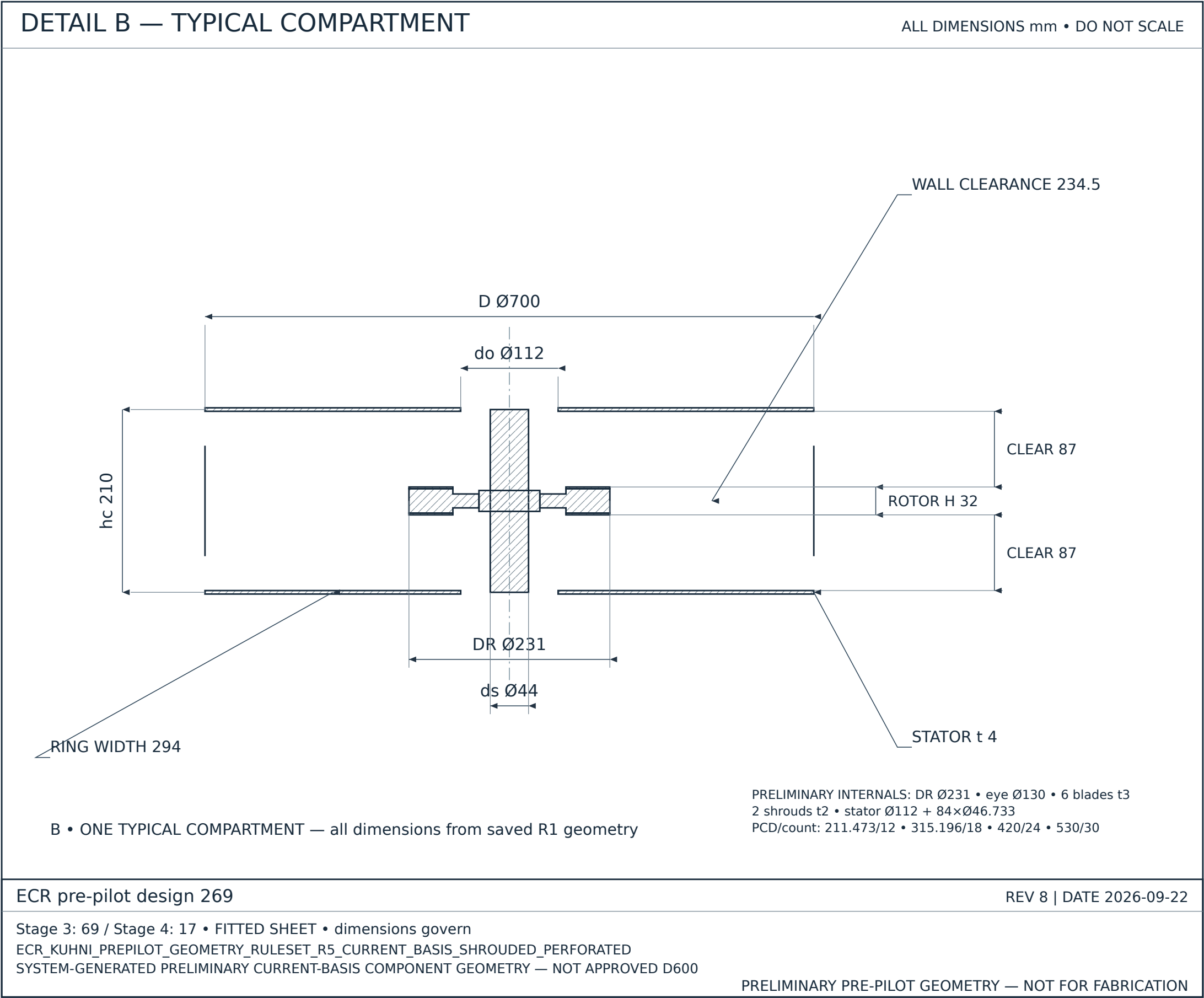
ECR pre-pilot design 269

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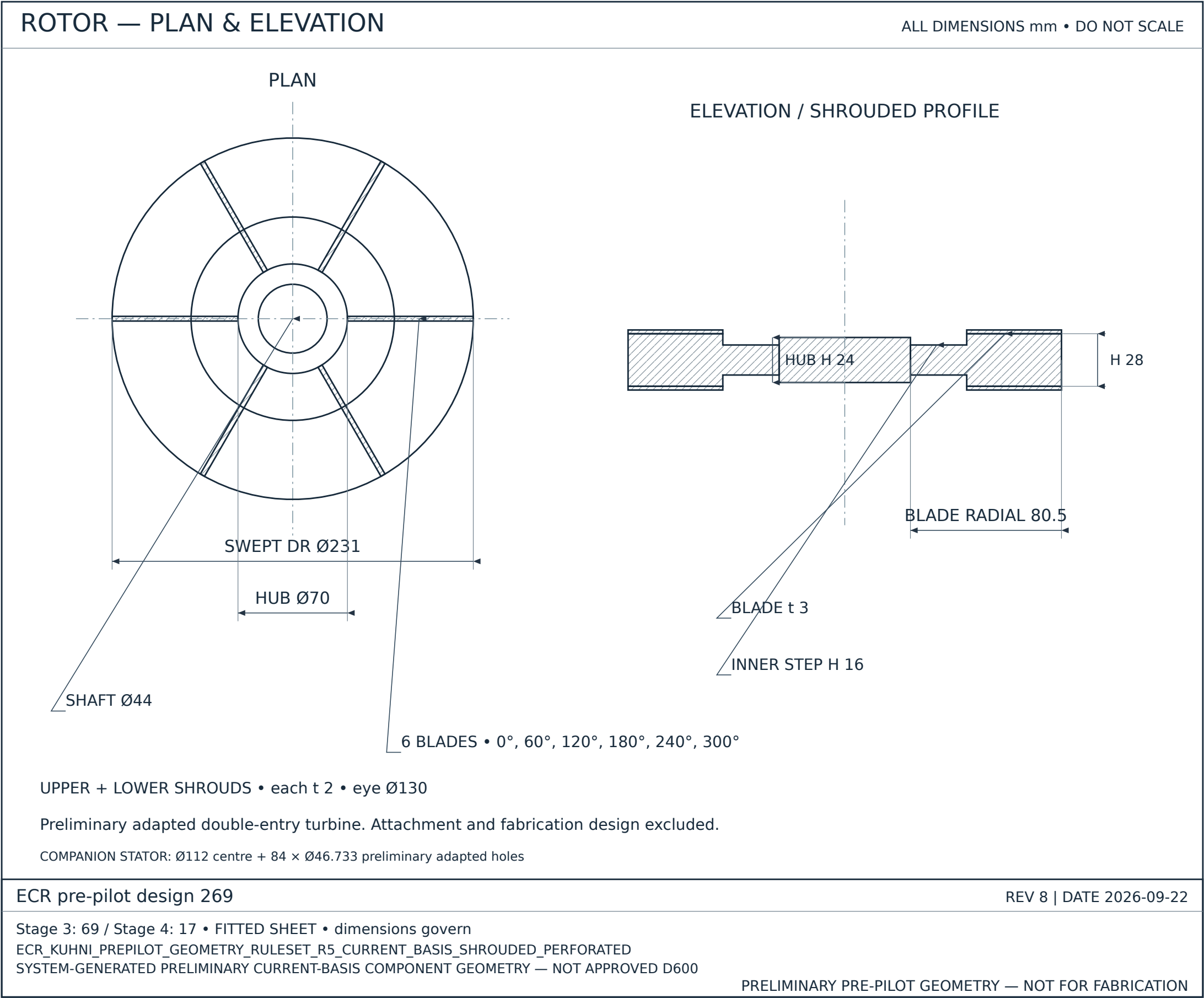
Stage 3: 69 / Stage 4: 17 • FITTED SHEET • dimensions govern
ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED
FROZEN ACTIVE ONLY — CURRENT ENDS PENDING

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION

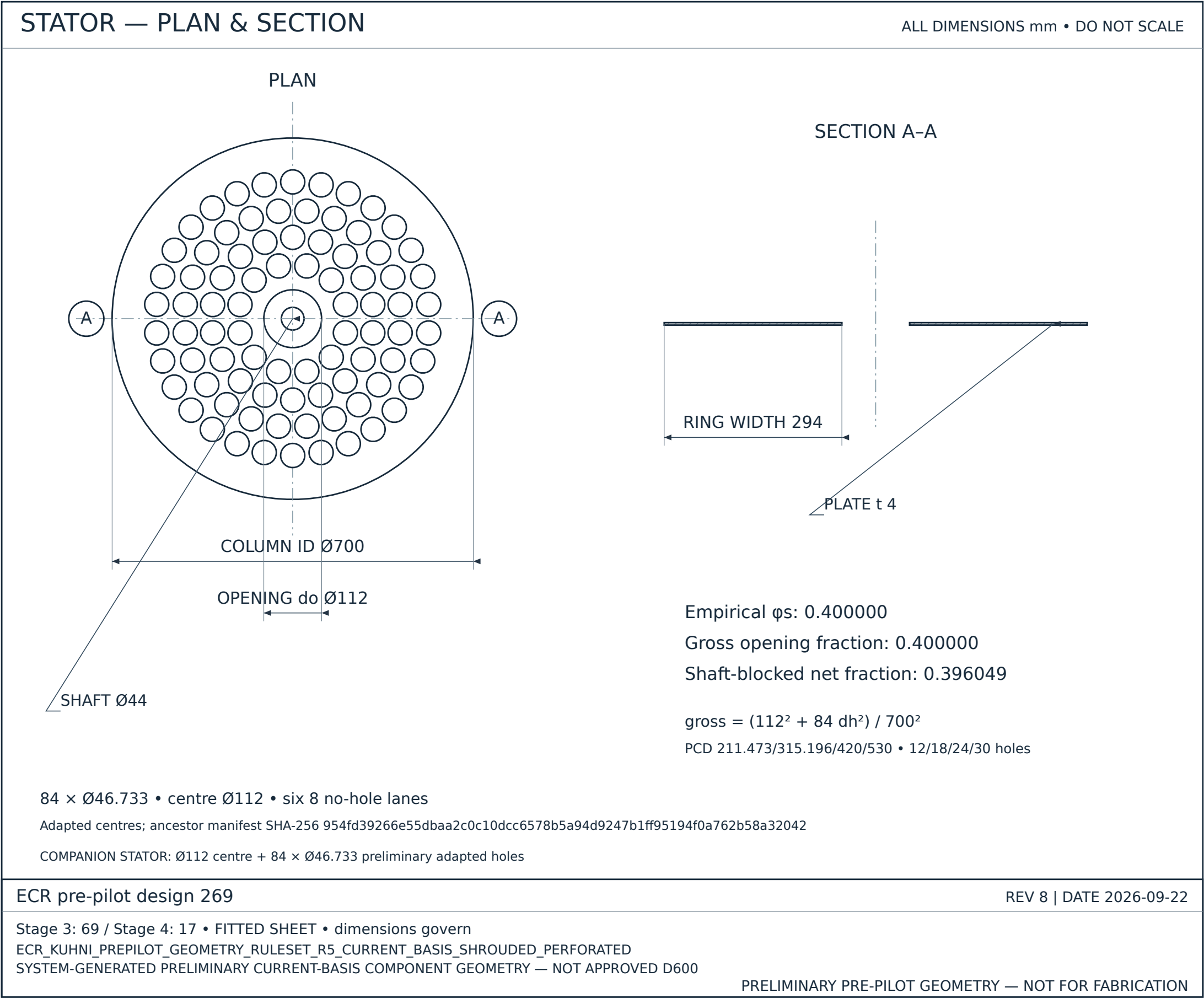
Typical compartment — frozen detail



Rotor — frozen detail



Stator — frozen detail



Compact traceability and audit handoff

Identity	Verified reference
Design / revision	269 / 8 (record 8)
Saved revision date	2026-09-22T05:52:17.942Z
Frozen active source SHA-256	d69bc153d009196c806639257001dd8abf0914f572e4d1809cbf5f440e078138
Current end source SHA-256	b7359f962cab45f79f5abdde27436e689a7c5434db99ebe3d7af31dd985a22fd
Stage 1 immutable hash	06bb9227b30f1f4c67b3ed557db34f61fe749037addf66163194dd46e2e98c5f
Geometry hash	ee377680bca1d61c1afb4fc0707fbf720de70f9f2ad04b7d8fc2485e5bcb3337
Frozen geometry ruleset	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R5_CURRENT_BASIS_SHROUDED_PERFORATED
Current end ruleset	ECR_END_SECTIONS_SYSTEM_AUTHORITY_PENDING_V3
Source status at read	CURRENT

The current report source is read in one consistent read-only transaction, with a single frozen revision hydration. A changed, stale, missing or invalid source fails closed; no historical end fallback is used.

Engineering Calculation & Audit Archive: complete saved Design Data report, including raw coordinates, full validation register, expanded Stage 3/4 reference evidence, provenance and historical end/nozzle reservations. It is independently readable from the verified frozen snapshot; it is not labelled current engineering authority.